Marc Kegel

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Personal

- Born on June 29, 1987 in Cologne.
- German citizen.

Research Interests

- **Geometric topology:** low dimensional topology, experimental mathematics, knot theory, 3-manifolds, smooth 4-manifolds, Dehn surgery, open books, Kirby calculus, Lefschetz fibrations, trisections and Heegaard splittings.
- **Differential topology:** contact and symplectic geometry, Legendrian and transverse knot theory, Engel structures, Morse theory, hyperbolic geometry.

Publications

Submitted articles:

(46) M. Golla and M. Kegel and Q. Ren, *Lens spaces that cannot be obtained by two integral surgeries*, `arXiv`:2610.16431, 21 pages.

(45) M. Kegel and Q. Ren, *Deciding strong quasipositivity and quasipositivity*, `arXiv`:2609.17341, 7 pages.

(44) M. Kegel, S. Y. Li, and Q. Ren, *Small undecidable groups and unrecognizable 4-manifolds*, `arXiv`:2609.10461, 14 pages.

(43) N. M. Dunfield, M. Kegel, S. Y. Li, and Q. Ren, *Trisection invariants of 4-manifolds are uncomputable*, `arXiv`:2608.27811, 6 pages.

(42) C.-S. Hsueh, M. Kegel, D. Suchodoll, and A. Thiele, *Legendrian simple knots not detected by Khovanov homology*, `arXiv`:2608.24353 , 4 pages.

(41) M. Kegel and M. Teragaito, *An infinite family of strongly invertible L-space knots without Khovanov thin surgery*, `arXiv`:2608.00864, 15 pages.

(40) C. Donatone, M. Kegel, L. Lewark, and P. Truöl, *The Conway knot has infinite concordance order*, `arXiv`:2606.30422, 14 pages.

(39) M. Kegel, E. Stenhede, and V. Vértesi, *A contact version of Kirby's theorem*, `arXiv`:2605.27126, 46 pages.

(38) M. Kegel and I. Nonino, *Transverse knots determined by their cyclic branched covers*, `arXiv`:2603.25805, 12 pages.

(37) S. Friedl, T. Hirsch, and M. Kegel, *The unsolvability of the homeomorphism problem*, arXiv:2603.23630, 7 pages.

(36) M. Kegel and J. Spreer, *The search for exotic knot traces*, arXiv:2603.22438, 23 pages.

(35) M. Kegel, I. Nonino, and M. Yadav, *Contact surgery distance*, arXiv:2512.14904, 24 pages.

(34) M. Kegel and M. Schmalian, *Links have no characterising slopes*, arXiv:2511.21506, 10 pages.

(33) M. Kegel and M. Schmalian, *Unique surgery descriptions along knots*, arXiv:2508.18521, 25 pages.

(32) M. Kegel and L. Piccirillo, *Knots that share four surgeries*, arXiv:2505.13168, 22 pages.

(31) R. Chatterjee and M. Kegel, *Contact surgery numbers of Sigma(2,3,11) and L(4m+3,4)*, arXiv:2404.18177, 30 pages.

(30) M. Kegel, N. Manikandan, L. Mousseau, and M. Silvero, *Khovanov homology of positive links and of L-space knots*, arXiv:2304.13613, 25 pages.

Articles accepted for publication:

(29) V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro, P. Truöl, L. Wakelin, *On the detection of knotted spheres by their traces in high dimensions*, arXiv:2511.07251, 46 pages, to appear in *Proceedings of the London Mathematical Society.*

(28) M. Kegel and P. Truöl, *Braid positive surgery descriptions*, arXiv:2512.14589, 7 pages, to appear in *Glasgow Journal of Mathematics.*

(27) T. Abe, M. Kegel, and N. Weiss, *Complexity of equal 0-surgeries*, arXiv:2401.06015, 32 pages, to appear in *New York J. Math.*

(26) K. Baker, M. Kegel, and D. McCoy, *Quasi-alternating surgeries*, arXiv:2409.09839, 28 pages, to appear in *Experimental Math.*

(25) S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löh, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perras, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, P. Truöl, M. Uschold, S. Veselá, M. Weiß, and M. von Wunsch-Rolshoven, *Algorithms in 4-manifold topology*, arXiv:2411.08775, 24 pages, to appear in *Algebr. Geom. Topol.*

(24) M. Kegel, L. Lewark, N. Manikandan, F. Misev, L. Mousseau, and M. Silvero, *On unknotting fibered positive knots and braids*, arXiv:2312.07339, 17 pages, to appear in *Ann. Sc. Norm. Super. Pisa Cl. Sci.*

Published articles:

(23) M. Kegel, A. Ray, J. Spreer, E. Thompson, and S. Tillmann, *On a volume invariant of 3-manifolds*, *Expositiones Math* **44** (2026).

(22) K. Baker, M. Kegel, and D. McCoy, *The search for alternating surgeries*, *J. Topol.* **19** (2026).

(21) K. Baker, M. Kegel, and K. Motegi, *Knots not detected by any trace*, *Int. Math. Res. Not. IMRN.* **2026** (2026).

(20) M. Kegel and M. Yadav, *Contact surgery numbers of projective spaces*, *Mediterr. J. Math.* **22** (2025).

(19) K. Baker, M. Kegel, and D. McCoy, *Two curious strongly invertible L-space knots*, *Adv. Math.* **473** (2025), 110287.

(18) M. Kegel and F. Schmäschke, *Trisecting a 4-dimensional book into three chapters*, *Geom. Dedicata* **218** (2024), Paper No. 86.

(17) R. Casals, J. Etnyre, and M. Kegel, *Stein traces and characterizing slopes*, *Math. Ann.* **389** (2024), 1053–1098.

(16) J. Etnyre, M. Kegel, and S. Onaran, *Contact surgery numbers*, *J. Symplectic Geom.* **21** (2023), 1255–1333.

(15) M. Kegel, *Non-isotopic transverse tori in Engel manifolds*, *Rev. Mat. Iberoamericana* **40** (2024), 43–56.

(14) K. Baker and M. Kegel, *Census L-space knots are braid positive, except for one that is not*, *Algebr. Geom. Topol.*, **24** (2024), 569–586.

(13) F. Ayaz, M. Kegel, and K. Mohnke, *The classification of surfaces via normal curves*, *Jahresber. Dtsch. Math.-Ver.*, **125** (2023), 121–134.

(12) C. Anderson, K. Baker, X. Gao, M. Kegel, K. Le, K. Miller, S. Onaran, G. Sangston, S. Tripp, A. Wood, and A. Wright, *L-space knots with tunnel number >1 by experiment*, *Experimental Math.*, **32** (2023), 600–614.

(11) M. Kegel and S. Onaran, *Contact surgery graphs*, *Bull. Aust. Math. Soc.*, **107** (2023), 146–157.

(10) S. Durst, M. Kegel, and J. Licata, *Rotation numbers and the Euler class in open books*, *Michigan Math. J.*, **70** (2021), 869–888.

(9) M. Kegel and C. Lange, *A Boothby–Wang theorem for Besse contact manifolds*, *Arnold Math. J.*, **7** (2021), 225–241.

(8) S. Durst, H. Geiges, J. Gonzalo, and M. Kegel, *Parallelisability of 3-manifolds via surgery*, *Expo. Math.*, **38** (2020), 131–137.

(7) M. Kegel, J. Schneider, and K. Zehmisch, *Symplectic dynamics and the* 3*-sphere*, *Israel J. Math.*, **235** (2020), 245–254.

(6) S. Durst, H. Geiges, and M. Kegel, *Handle homology of manifolds*, *Topology Appl.*, **256** (2019), 113–127.

(5) M. Kegel, *Cosmetic contact surgeries along transverse knots and the knot complement problem*, *Topology Appl.*, **256** (2019), 46–59.

(4) S. Durst and M. Kegel, *Computing rotation numbers in open books*, *J. Gökova Geom. Topol. GGT* **12** (2018), 71–92.

(3) M. Kegel, *The Legendrian knot complement problem*, *J. Knot Theory Ramifications*, **27** (2018), 1850067, 36 pages.

(2) S. Durst and M. Kegel, *Computing rotation and self-linking numbers in contact surgery diagrams*, *Acta Math. Hungar.* **150** (2016), 524–540.
Erratum, *Acta Math. Hungar.* **153** (2017), 537.

(1) S. Durst, M. Kegel, and M. Klukas, *Computing the Thurston–Bennequin invariant in open books*, *Acta Math. Hungar.* **150** (2016), 441–455.

Other documents

Non-refereed contributions:

(7) M. Kegel, *A complete set of contact Kirby moves*, Oberwolfach Reports, to appear.

(6) V. Bais, A. Barbensi, B. Burton, D. Celoria, C.-S. Hsueh, M. Kegel, J. Licata, A. Merz, J. Spreer, A. Thiele, and L. Tobin, *Models for random knots*, Oberwolfach Reports **43** (2024), 5 pages.

(5) M. Kegel and M. Silvero, *Khovanov homology of positive knots*, ICERM Highlights (2023), 3 pages.

(4) M. Kegel, J. Licata, and A. Ray, *Discussions on knot theory in general 3-manifolds*, Oberwolfach Reports **42** (2021), 4 pages.

Theses:

(3) M. Kegel, *Legendrian knots in surgery diagrams and the knot complement problem*, Doktorarbeit (Ph.D. thesis), Universität zu Köln, (January 2017), 123 pages.

(2) M. Kegel, *Symplektisches Füllen von Torusbündeln (Symplectic Fillings of Torus Bundles)*, Masterarbeit (Master thesis), Universität zu Köln (August 2014), 176 pages.

(1) M. Kegel, *Kontakt-Dehn-Chirurgie entlang Legendre-Knoten (Contact Dehn Surgery along Legendrian Knots)*, Bachelorarbeit (Bachelor thesis), Universität zu Köln (September 2011), 61 pages.

Employment

- **September 2025–August 2030:** Ramón y Cajal fellowship at University of Sevilla.
- **June 2025 – August 2025:** Position of the DFG (German research foundation), *Eigene Stelle*, at the Humboldt University Berlin.
- **September 2024 – May 2025:** Postdoc at the Humboldt University Berlin in the working group of Prof. Chris Wendl.
- **April 2024 – September 2024:** Substitute professorship at the Ruhr-University Bochum.
- **September 2023 – March 2024:** Substitute professorship at the Ruprecht-Karls-University Heidelberg.
- **July 2023 – August 2024:** Postdoc at the Humboldt University Berlin in the working group of Prof. Chris Wendl.
- **February 2019 – June 2023:** Visiting lecturer (Gastdozent) at the Humboldt University Berlin in the working group of Prof. Chris Wendl.
- **August 2018 – January 2019:** BMS substitute professor at the Humboldt University Berlin in the working group of Prof. Chris Wendl.
- **July 2015 – July 2018:** Teaching assistant (Wissenschaftlicher Mitarbeiter) at the University of Cologne in the working group of Prof. Hansjörg Geiges.

Fellowships and Grants

- **2025 – 2030:** Ramón y Cajal fellowship at the Spanish research ministry, (total funding granted for 5 years 321.800 €).
- **2025-2026:** DFG *Eigene Stelle* 2025, at the German Research Foundation, (total funding granted for 2 years 225.000 €).
- **2025-2026:** Grant from the University of Sevilla (VII Plan Propio de Investigación y Transferencia) for conference and workshop organization, (total funding granted for 2 years 18.000 €).
- **May – July 2025:** Postdoc position for Monika Yadav at the HU Berlin, via the *Math+ Hanna Neumann Fellowship*, (total funding granted 10.000 €).
- **November – December 2024:** Postdoc position for Monika Yadav at the HU Berlin, via the WINS Postdoctoral Fellowship, (total funding granted 4.000 €).
- **March 2024:** BMS (Berlin Mathematical School) faculty member.
- **August 2023 – August 2026:** Ph.D. position for Chun-Sheng Hsueh funded by the Claussen-Simon Stiftung.
- **February – May 2022:** Semester program on braids at ICERM, Brown University, Providence.
- **August 2020:** Oberwolfach research fellow for one month at the MFO (Mathematisches Forschungsinstitut Oberwolfach).
- **August 2018 – January 2019:** BMS substitute professor at the Humboldt University Berlin via the Berlin mathematical school.
- **September 2014 – June 2015:** Fellow of the DFG (German Research Foundation) Graduiertenkolleg 1269 "Global Structures in Geometry and Analysis" at the University of Cologne.

Education

- **January 2017: Ph.D.** in mathematics (Dr. rer. nat.) at University of Cologne,
  "Legendrian knots in surgery diagrams and the knot complement problem", 123 pages,
  Advisor: Prof. Hansjörg Geiges,
  Grade: magna cum laude (1,0).
- **August 2014: Master** in mathematics (M.Sc.) (Minor: physics) at University of Cologne,
  Grade: sehr gut (1,1),
  "Symplektisches Füllen von Torusbündeln" (Symplectic Filling of Torus Bundles), 176 pages,
  Thesis advisor: Prof. Hansjörg Geiges,
  Thesis grade: 1,0.
- **September 2011: Bachelor** in mathematics (B.Sc.) (Minor: physics) at University of Cologne
  Grade: sehr gut (1,3),
  "Kontakt-Dehn-Chirurgie entlang Legendre-Knoten" (Contact Dehn Surgery along Legendrian Knots), 61 pages,
  Thesis advisor: Prof. Hansjörg Geiges,
  Thesis grade: 1,0.
- **June 2007: Allgemeine Hochschulreife** at Gymnasium Köln Nippes.

Research Talks

Invited talks at conferences or workshops:

(20) *The Conway knot has infinite concordance order*, July 2026, Trisectors Workshop 2026, Western Washington University (United States).

(19) *A complete set of contact Kirby moves*, July 2026, Workshop on low-dimensional topology, Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach (Germany).

(18) *The search for exotic knot traces*, June 2026, Berlin-Brandenburg Workshop VII: Knot Theory and its Application, HU Berlin (Germany).

(17) *Unique surgery descriptions along knots*, January 2026, Meeting of the Royal Spanish Mathematical Society, Alicante (Spain).

(16) *Unique surgery descriptions along knots*, October 2025, Conference on Trisections and related topics, CIRM, Marseille (France).

(15) *Exotic knot traces via Khovanov homology*, July 2025, Workshop on Khovanov homology and exotic 4-manifolds, Erdös institute, Budapest (Hungary).

(14) *Adjunction inequalities in Khovanov homology*, July 2025, Workshop on Khovanov homology and exotic 4-manifolds, Erdös institute, Budapest (Hungary).

(13) *The search for exotic knot traces*, February 2025, BMS Days Conference, TU Berlin (Germany).

(12) *The search for exotic knot traces*, December 2024, Workshop on Prospects in Differential Geometry, Hannover (Germany).

(11) *Characterizing and non-characterizing knots by* 3*-manifolds*, November 2024, Workshop on low-dimensional topology, University of Kyoto (Japan).

(10) *Characterizing and non-characterizing knots by* 3*-manifolds*, July 2024, Braids Reunion Workshop, ICERM, Brown University, Providence (United States).

(9) *Contact surgery numbers*, February 2024, CAST Conference on contact and symplectic topology, University of Bochum (Germany).

(8) *Characterizing and non-characterizing knots by* 3*-manifolds*, September 2023, Conference on low-dimensional topology, TIFR Mumbai (India).

(7) *Characterizing and non-characterizing knots by* 3*-manifolds*, September 2023, Swiss Knots, Conference on knot theory and low dimensional topology in Switzerland, University of Regensburg (Germany).

(6) *Stein traces*, October 2022, BACH-Seminar, University of Cologne (Germany).

(5) *Census L-space knots are braid positive, except for one that is not*, February 2022, Braids in representation theory and algebraic combinatorics, ICERM, Brown University, Providence (United States).

(4) *Characterizing slopes for Legendrian knots*, July 2020, Minisymposium: Geometric analysis and low-dimensional topology at 8th European congress of mathematicians, Portoroz (online).

(3) *Contact surgery numbers*, January 2021, Workshop on 3-dimensional contact topology, Ankara (online).

(2) *The knot complement problem for Legendrian and transverse knots*, May 2018, Symplectix Seminar Paris (France).

(1) *The knot complement problem for Legendrian and transverse knots*, January 2018, Berlin-Hamburg Seminar zur symplektischen Geometrie, Humboldt-Universität zu Berlin (Germany).

Minicourses at schools or workshops:

(3) *Describing* 3*-manifolds as branched covers* (3 talks), November 2025, Workshop on branched covers, Sevilla (Spain).

(2) *The search for exotic knot traces* (2 talks), November 2023, Dutch Differential Topology and Geometry seminar, Utrecht (Netherlands).

(1) *Minicourse on* 4*-manifolds and Kirby calculus* (2 talks), September 2023, Summer school on low-dimensional topology, IISER Pune (India).

Invited talks at seminars:

(35) *The Conway knot has infinite concordance order*, July 2026, University of Berkeley (United States).

(34) *A complete set of contact Kirby moves*, June 2026, Ruhr University Bochum (Germany).

(33) *Algorithms in contact topology*, June 2026, Ruhr University Bochum (Germany).

(32) *Algorithms in* 4*-manifold topology*, June 2026, University of Cologne (Germany).

(31) *The search for exotic knot traces*, April 2026, ETH Zürich (Switzerland).

(30) *Unique surgery descriptions along knots*, February 2026, University of Glasgow (UK).

(29) *Characterizing and non-characterizing knots by* 3*-manifolds*, February 2026, University of Oxford (UK).

(28) *Unique surgery descriptions along knots*, October 2025, [K-OS] Knot Online Seminar (online).

(27) *Dehn surgery and exotic 4-manifolds*, September 2025, University of Porto (Portugal).

(26) *The search for exotic knot traces*, December 2024, Oberseminar Geometry and Topology, University of Cologne (Germany).

(25) *Characterizing and non-characterizing knots by* 3*-manifolds*, October 2024, Max-Planck-Institut für Mathematik, Bonn (Germany).

(24) *The search for exotic knot traces*, May 2024, Geometry Seminar, Heidelberg University (Germany).

(23) *Characterizing and non-characterizing knots by* 3*-manifolds*, October 2023, Topology Seminar, University of Texas at Austin (United States).

(22) *Khovanov homology of positive links and of L-space knots*, May 2023, University of Trieste (Italy).

(21) *The search for exotic knot traces*, March 2023, Seminar of Algebra, University of Seville (Spain).

(20) *Hyperbolic L-space knots*, February 2023, University of Prag (Czech Republic).

(19) *Hyperbolic L-space knots*, February 2023, Koc University, Istanbul (Turkey).

(18) *The search for alternating and quasi-alternating surgeries*, November 2022, Oberseminar Geometrische Analysis, University of Regensburg (Germany).

(17) *Transverse tori in Engel manifolds*, October 2022, Symplectic Topology Seminar, University of Cologne (Germany).

(16) *The search for alternating and quasi-alternating surgeries*, September 2022, Max-Planck-Institut für Mathematik, Bonn (Germany).

(15) *Non-isotopic transverse tori in Engel manifolds*, July 2022, Geometry seminar, University of Munich (Germany).

(14) *Stein traces*, May 2022, Freemath seminar (online).

(13) *The search for alternating and quasi-alternating surgeries*, March 2022, Computational seminar for the semester program on braids at ICERM, Brown University, Providence (United States).

(12) *Stein traces*, February 2022, Massachusetts Institute of Technology (United States).

(11) *Stein traces*, January 2022, Shanghai Jiao Tong University (China).

(10) *Characterizing slopes for Legendrian knots*, January 2021, Göttingen topology and geometry seminar (online).

(9) *Contact surgery numbers*, April 2020, Regensburg working group (online).

(8) *Constructing symplectic structures from Engel manifolds*, December 2019, Symplectic geometry seminar Heidelberg (Germany).

(7) *Morse structures on open books and the Euler class of contact structures*, December 2019, Geometry Seminar Heidelberg (Germany).

(6) *The knot complement problem for Legendrian and transverse knots*, January 2018, Hauptseminar Symplektische Geometrie, Universität Heidelberg (Germany).

(5) *The knot complement problem for Legendrian and transverse knots*, January 2018, Berlin-Hamburg Seminar zur symplektischen Geometrie, Humboldt-Universität zu Berlin (Germany).

(4) *The knot complement problem for Legendrian and transverse knots*, December 2017, Oberseminar Topologie, Bergische Universität Wuppertal (Germany).

(3) *The knot complement problem for Legendrian and transverse knots*, October 2017, Oberseminar Symplektische Geometrie, Westfälische Wilhelms Universität Münster (Germany).

(2) *The knot complement problem for Legendrian and transverse knots*, June 2017, Geometry seminar, University of Munich (Germany).

(1) *The Legendrian knot complement problem*, June 2016, LKS-Seminar, University Regensburg (Germany).

Contributed talks:

(12) *Knot traces in high dimensions*, July 2026, Workshop on low-dimensional topology, Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach (Germany).

(11) *Unique surgery descriptions along knots*, June 2026, Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions, Budapest (Hungary).

(10) *Census L-space knots are braid positive, except for one that is not*, February 2023, Winterbraids XII, Tours (France).

(9) *Khovanov homology of positive fibered links and L-space knots*, January 2023, Workshop on morphisms in low-dimensional topology, Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach (Germany).

(8) *Open books on 4-manifolds*, June 2022, Conference on surfaces in 4-manifolds, Le Croisic (France).

(7) *Stein traces*, March 2022, Braids in Symplectic and Algebraic Geometry, ICERM, Brown University, Providence (United States).

(6) *Algorithms with fibered links and closures of positive braids*, March 2022, Post Doc/Graduate Student Seminar for the semester program on braids at ICERM, Brown University, Providence (United States).

(5) *Contact surgery numbers*, April 2020, Trends in low dimensional topology (online).

(4) *Taut foliations and tight contact structures* (2 talks), October 2019, Workshop on Foliations, Regensburg (Germany).

(3) *The knot complement problem for Legendrian and transverse knots*, February 2018, Winter Braids VIII, CIRM, Marseille (France).

(2) *Lagrangian projections in contact* 3*-manifolds*, September 2017, SFB Retreat 2017, Tagungszentrum Maria in der Aue (Germany).

(1) *The knot complement problem for Legendrian and transverse knots*, October 2016, Conference on 4-manifolds and knot concordance, Max Planck Institute for Mathematics, Bonn (Germany).

Local seminar talks:

(23) *Applications of Heegaard Floer homology*, December 2024, Symplectic geometry seminar, Humboldt University Berlin.

(22) *Contact Kirby moves*, November 2024, Working group seminar on geometric topology, Humboldt University Berlin.

(21) *Detecting fibered links via Reeb dynamics*, July 2024, ACDC Seminar, Ruhr-University Bochum.

(20) *Stein traces*, November 2023, Symplectic geometry seminar, Heidelberg University.

(19) *Transverse tori in Engel manifolds*, October 2022, Symplectic geometry seminar, Humboldt University Berlin.

(18) *Stein traces*, November 2021, Symplectic geometry seminar, Humboldt University Berlin.

(17) *Heegaard Floer homology and the isomorphism with embedded contact homology I - Introduction to Heegaard Floer homology and Lipshitz's cylindrical reformulation*, February 2021, Symplectic geometry seminar, Humboldt University Berlin.

(16) *Surgery along Chekanov's knots*, July 2020, Symplectic geometry seminar, Humboldt University Berlin.

(15) *Open questions about open books*, April 2019, Symplectic geometry seminar, Humboldt University Berlin.

(14) *Parallelizability of 3-manifolds via surgery* (2 talks), October 2018, Symplectic geometry seminar, Humboldt University Berlin.

(13) *Unit tangent bundles of orbifolds*, October 2017, Symplectic topology seminar, University of Cologne.

(12) *Normal curves on surfaces*, May 2017, Symplectic topology seminar, University of Cologne.

(11) *Legendrian knots in surgery diagrams and the knot complement problem*, January 2017, Disputation, University of Cologne.

(10) *Contact cabling*, July 2016, Symplectic topology seminar, University of Cologne.

(9) *Complements of transverse knots*, April 2016, Symplectic topology seminar, University of Cologne.

(8) *Surgery descriptions of Legendrian knots*, April 2016, Symplectic topology seminar, University of Cologne.

(7) *Orbifolds and coverings*, January 2016, Symplectic topology seminar, University of Cologne.

(6) *Computation of the Thurston–Bennequin invariant in surgery diagrams*, October 2015, Symplectic topology seminar, University of Cologne.

(5) *Legendrian knots and their complements*, July 2015, Symplectic topology seminar, University of Cologne.

(4) *Stable Hamiltonian structures*, April 2015, Symplectic topology seminar, University of Cologne.

(3) *Symplectic fillability of torus bundles*, October 2014, Symplectic topology seminar, University of Cologne.

(2) *The topology of Lefschetz fibrations* (2 talks), October 2014, Symplectic topology seminar, University of Cologne.

(1) *Contact surgery along Legendrian links* (2 talks), June 2011, Symplectic topology seminar, University of Cologne.

Research stays

- **July 2026:** *University of Berkeley*, United States, 1 week. Host: Siddhi Krishna
- **July 2026:** *Humboldt University Berlin*, Germany, 1 week. Host: Chris Wendl
- **June 2026:** *University of Cologne*, Germany, 1 week. Host: Hanjörg Geiges
- **June 2026:** *Max-Planck-Institut für Mathematik*, Bonn, Germany, 1 week. Host: Rima Chatterjee
- **April 2026:** *ETH Zürich*, Switzerland, 1 week. Host: Lukas Lewark
- **February 2026:** *University of Glasgow*, UK, 1 week. Host: Paula Truöl
- **February 2026:** *University of Oxford*, UK, 1 week. Host: Misha Schmalian
- **April 2025:** *University Seville*, Spain, 1 week. Host: Marithania Silvero
- **November 2024:** *Kyoto*, Japan, 1 week. Host: Tetsuya Ito
- **July 2024:** *Montreal*, Canada, 1 week. Host: Duncan McCoy
- **November 2023:** *Universiteit Utrecht*, Netherlands, 1 week. Host: Álvaro del Pino Gómez
- **October 2023:** *University of Texas at Austin*, United States, 1 week. Host: Lisa Piccirillo
- **September 2023:** *IISER Pune*, India, 2 weeks. Host: Tejas Kalelkar
- **March 2023:** *University Seville*, Spain, 1 week. Host: Marithania Silvero
- **October 2022:** *University of Cologne*, Germany, 1 week. Host: Hanjörg Geiges
- **September 2022:** *University of Vienna*, Austria, 1 week. Host: Vera Vértesi
- **September 2022:** *Max-Planck-Institut für Mathematik*, Bonn, Germany, 1 week. Host: Arunima Ray
- **March 2022:** *University of Miami*, USA, 1 week. Host: Kenneth L. Baker

- **February – April 2022:** *ICERM, Brown University*, Providence, USA, 3 months. As part of the semester program on "Braids".
- **August 2020:** *Mathematisches Forschungsinstitut Oberwolfach*, Germany, 1 month. As Oberwolfach research fellow.
- **February 2018:** *CIRM, Centre International de Rencontres Mathématiques*, Marseille, France, 2 weeks. For participating in the conferences "Winter Braids VIII" and "Knotted Embeddings in Dimension 3 and 4".
- **November 2017:** *Westfälische Wilhelms Universität Münster*, Germany, 3 days. Host: Kai Zehmisch

Organization of research events

- **September 2026** Workshop on Algorithms in Contact Topology, University Sevilla, planned.
- **February 2026** Workshop on Geometric Topology, University Sevilla.
- **December 2025** Workshop on surfaces in 4-manifolds, University Regensburg (together with S. Friedl, M. Powell, and A. Ray).
- **January 2026** Symposium on Low-dimensional Topology at the Indo-European Conference on Mathematics, Pune, India (together with T. Kalelkar).
- **December 2025** Workshop on surfaces in 4-manifolds, University Regensburg (together with S. Friedl, M. Powell, and A. Ray).
- **November 2025** Workshop on branched covers, University Sevilla (together with M. Silvero).
- **April 2025** Berlin-Brandenburg Workshop V: Knot Theory and its Applications, Humboldt University Berlin (together with T. Andriamanalina, C.-S. Hsueh, and N. Manikandan).
- **November 2024** Berlin-Brandenburg Workshop IV: Knot Theory and its Applications, Humboldt University Berlin (together with T. Andriamanalina, C.-S. Hsueh, and N. Manikandan).
- **September 2024** Workshop on 4-manifolds and algorithms, University Regensburg (together with S. Friedl).
- **12 July 2024** Berlin-Brandenburg Workshop III: Knot Theory and its Applications, University Potsdam (together with T. Andriamanalina and N. Manikandan).
- **17 November 2023** Berlin-Brandenburg Workshop II: Knot Theory and its Applications, Freie Universität Berlin (together with T. Andriamanalina and J. Spreer).
- **14 July 2023** Berlin-Brandenburg Workshop I: Knot Theory and its Applications, Humboldt University Berlin (together with C.-S. Hsueh and N. Manikandan).
- **September 2022:** *DMV-Jahrestagung 2022*, Section S06: Topology and Geometry, Freie Universität Berlin (Germany). Organizer (together with N. Bottman, M. Land, and L. Lewark).
- **Summer 2021 – 2023** Student research seminar on the SnapPy census knots, weekly research seminar for students, Humboldt University Berlin.

Teaching

Ruhr University Bochum:

- **Summer term 2024:** Lecture ”Geometric Topology’ (weekly 4 hours lecture + 2 hours exercise session).
- **Summer term 2024:** Seminar on contact geometry (weekly 2 hours).

Ruprecht-Karls-University Heidelberg:

- **Winter term 2023/24:** Lecture ”Contact geometry” (weekly 4 hours lecture + 2 hours exercise session, in total: 96 hours teaching).
- **Winter term 2023/24:** Seminar on hyperbolic knot theory (weekly 2 hours, in total: 32 hours teaching).
- **Winter term 2023/24:** RTG-Lecture ”Handling the Poincaré conjecture” (2 hours lecture every other week, in total: 12 hours teaching).

Humboldt University Berlin:

- **Summer term 2025:** Lecture "Topologie I" (weekly 4 hours lecture, in total: 64 hours teaching).
- **Winter term 2024/25:** Lecture "Funktionentheorie" (weekly 4 hours lecture, in total: 12 hours teaching).
- **Winter term 2024/25:** Exercise session for the lecture "Topologie II" (weekly 2 hours, in total: 32 hours teaching).
- **Summer term 2023:** Seminar on hyperbolic knot theory (weekly 2 hours, in total: 28 hours teaching).
- **Winter term 2022/23:** Lecture "Differential Geometry I: Curves and surfaces" (weekly 4 hours lecture, in total: 64 hours teaching).
- **Summer term 2022:** Exercise session for "Differential Geometry II" (weekly 2 hours, in total: 28 hours teaching).
- **Winter term 2021/22:** Lecture "Topology II" (weekly 4 hours lecture, in total: 64 hours teaching).
- **Summer 2021** – Student research seminar on the SnapPy census knots, weekly research seminar for students. (Research seminar for students, no official teaching, but weekly meetings.)
- **Summer term 2021:** Lecture "4-Manifolds and Kirby calculus" (weekly 2 hours lecture + 1 hour exercise session, in total: 42 hours teaching).
- **Winter term 2020/21:** Lecture "Topology II" (weekly 4 hours lecture + 2 hours exercise session, in total: 96 hours teaching).
- **Winter term 2020/21:** "Seminar on selected topics in algebraic and differential topology" (weekly 2 hours, in total: 32 hours teaching).
- **Summer term 2020:** Lecture "Topology of 3-Manifolds" (weekly 2 hours lecture + 1 hour exercise session, in total: 42 hours teaching).
- **Winter term 2019/20:** Lecture "Topology II" (weekly 4 hours lecture + 2 hours exercise session, in total: 96 hours teaching).

- **Summer term 2019:** Lecture "Topologie I" (weekly 4 hours lecture + 2 hours exercise session, in total: 84 hours teaching).
- **Winter term 2018/19:** "Seminar zur Differentialtopologie" (weekly 2 hours, in total: 32 hours teaching).
- **Winter term 2018/19:** Exercise session for the lecture "Geometrie" by Prof. Klaus Mohnke (weekly 2 hours, in total: 32 hours teaching).

University of Cologne:

- **Summer term 2018:** Lecture "Kirby-Kalkül" (weekly 2 hours lecture + 2 hours exercise session, in total: 56 hours teaching).
- **Winter term 2017/18:** Teaching Assistant for the lecture "Elementare Differentialgeometrie" by Prof. Silvia Sabatini.
- **Summer term 2017:** Teaching Assistant and exercise session for the lecture "Topologie" by Prof. Hansjörg Geiges (weekly 2 hours, in total: 30 hours teaching).
- **Winter term 2016/17:** Teaching Assistant and exercise session for the lecture "Symplectic Geometry" by Dr. Milena Pabiniak (weekly 2 hours, in total: 30 hours teaching).
- **Summer term 2016:** Teaching Assistant and exercise session for the lecture "Differential Geometry" by Dr. Milena Pabiniak (weekly 2 hours, in total: 30 hours teaching).
- **Winter term 2015/16:** Teaching Assistant for "Seminar über Topologie" by Prof. Hansjörg Geiges (weekly 2 hours, in total: 30 hours teaching).
- **2010–2014:** Tutor for mathematical lectures, including Mathematics I and II, Differential Equations, Complex Analysis, Geometry of Celestial Mechanics, Topology and Dynamical Systems, Surfaces and Elementary Differential Geometry (weekly 2-4 hours).
- **2008–2011:** Teacher training for mathematics and physics at University of Cologne, including courses in pedagogy and concerning the principles of teaching mathematics and physics.
- **September 2009:** Internship at Erich Kästner-Gymnasium Köln Niehl, 2 weeks.

Postdocs mentored:

- Monika Yadav, Humboldt University Berlin, May – July 2024, with funding from the *Hannah–Neumann fellowship via the MATH+.*
- Monika Yadav, Humboldt University Berlin, November – December 2024, with funding from the *WINS Postdoctoral fellowship.*

Supervised Ph.D. theses:

- Naageswaran Manikandan, (cosupervision with Chris Wendl) Humboldt University Berlin, October 2021 – September 2025, with funding from the *Berlin Mathematical School*, now postdoc at the MPI Bonn.
- Chun-Sheng Hsueh, Humboldt University Berlin, September 2023 – September 2026, with funding from the *Claussen-Simon-foundation.*
- David Suchudoll, (cosupervision with Klaus Mohnke) Humboldt University Berlin, May 2022 – January 2027.

BMS student mentor:

- Annika Thiele, Humboldt University Berlin, since April 2024.

BMS qualifying exam:

- Laurenz Upmeier zu Belzen, 2023, Humboldt University Berlin.
- Li Li, 2021, Humboldt University Berlin.

Supervised master theses:

- Farid A. Azar León, *Torsion in the Khovanov homology of* 3*-braids*, Humboldt University Berlin, in progress.
- Cristina Gómez Cirera, *Lens space surgeries and the Berge conjecture*, 2026, Universidad de Sevilla.
- Frank Selensky, *Obstruction classes of contact structures*, 2026, Humboldt University Berlin.
- Léo Mousseau, *The L-space conjecture*, 2025, Humboldt University Berlin.
- Sarah Pestkowski, *Ein optimales Papier-Möbius-Band*, 2025, Humboldt University Berlin.
- Chun-Sheng Hsueh, *Kirby diagrams of* 4*-dimensional open books*, 2023, Humboldt University Berlin. Published as: C.-S. Hsueh, *Kirby diagrams of 4-dimensional open books*, arXiv:2306.16942, submitted.
- David Suchodoll: *The Thurston–Bennequin invariant in branched covers*, 2022, Humboldt University Berlin.

Master theses, second evaluator:

- Levent Kotan, *Operads*, Humboldt University Berlin, 2025, Freie University Berlin.
- Laurenz Upmeier zu Belzen, *Symplectic Fillability of Planar Open Books*, 2023, Humboldt University Berlin.

Supervised Bachelor theses:

- Arthur Berns, *The 6-theorem*, 2024, Humboldt University Berlin.
- Annika Thiele, *Algorithmic detections of fibered and positive knots*, 2023, Humboldt University Berlin. This thesis was awarded with a price from the *Berlin Mathematical Society.*
- Luis Kristic, *Hyperbolic knot theory and the A-polynomial*, 2023, Humboldt University Berlin.
- Frank Selensky, *Chern classes of high dimensional contact structures via open books*, 2023, Humboldt University Berlin. Currently under preparation for publication.
- Léo Mousseau, *Algorithms for computing the slice* 4*-genus and the unknotting number*, 2023, Humboldt University Berlin.
- Pau Punset, *Fibered links and orientations*, 2023, Universidad de Barcelona (as external supervisor).
- Fetih Ayaz, *Classification of surfaces via normal curves*, 2021, Humboldt University Berlin. Published as: F. Ayaz, M. Kegel, and K. Mohnke, *The classification of surfaces via normal curves*, *Jahresber. Dtsch. Math.-Ver.*
- Max Huneshagen, *Das HOMFLYPT-Polynom*, 2020, Humboldt University Berlin.

- Léo Duc, *Regular homotopy classes of Legendrian immersions*, 2019, Humboldt University Berlin.
- Paulina Bock de Barillas, *Der Satz von Whitney–Graustein, ein klassischer und ein kontaktgeometrischer Beweis*, 2019, Humboldt University Berlin.
- Bernhard Albach, *Reidemeisterbewegungen von Legendre-, transversalen und glatten Verschlingungen*, 2018, (supervised together with Hansjörg Geiges), University of Cologne.

Bachelor theses, second evaluator:

- Vincent Woltmann, *Proving de Rham's theorem via the Cech-de Rham double complex*, 2025, Humboldt University Berlin.
- Florian Buck, *Kontaktstrukturen auf Brieskorn-Mannigfaltigkeiten*, 2004, Ruhr University Bochum.
- Joachim Albrecht, *Hyperbolic metrics on surfaces with finitely many punctures*, 2024, Humboldt University Berlin.
- Leonard Vetter, *Topological recursion and open-closed topological quantum field theories*, 2023, Humboldt University Berlin.
- Maurice Parzonka, *Das Lemma von Sperner und seine Anwendung in einem Beweis des Brouwerschen Fixpunktsatzes*, 2023, Humboldt University Berlin.
- Luisa Gerlach, *Satz von Markov - von Knoten und Zöpfen*, 2022, Humboldt University Berlin.
- Jan Philipp Bohl, *Existence of contact structures on 3-manifolds*, 2021, Humboldt University Berlin.
- Jim Ye, *Hurwitz-Zahlen und ihre Berechnungsmethoden*, 2021, Humboldt University Berlin.
- Marie Christin Schmidtlein, *Symplectic fillings and Lefschetz fibrations in dimension four*, 2020, Humboldt University Berlin.
- Laurenz Upmeier zu Belzen, *Milnor construction of universal bundles*, 2019, Humboldt University Berlin.

Supervised final projects for the lecture Kirby-Kalkül (SS 2018):

- Bernhard Albach, *Blätterungen von 3-Mannigfaltigkeiten*, University of Cologne.
- Franziska Frede, *Nicht-destabilisierbare Heegaard-Zerlegungen*, University of Cologne.
- Konstantin Müller, *Heegaard- und Henkelzerlegungen von 3-Mannigfaltigkeiten mit Rand*, University of Cologne.
- Laura Maria Poreschack, *Homologie und Henkelzerlegungen*, University of Cologne.
- Lennart Struth, *Hakens Lemma*, University of Cologne.

Service

- Reviewer for *MathSciNet* and *Zentralblatt MATH*.
- Referee and quick opinions for *Bulletin of the Australian Mathematical Society*, *Bulletin of the Brazilian Mathematical Society*, *Bulletin of the London Mathematical Society*, *Canadian Mathematical Bulletin*, *International Mathematics Research Notices*, *Journal of Symplectic Geometry*, *K3: Kirby's problem list*, *Mediterranean Journal of Mathematics, Memoirs of the AMS*, *New York Journal of Mathematics*, *Michigan Mathematical Journal*, *Proceedings of the Edinburgh Mathematical Society*, *Transactions of the American Mathematical Society* and *Topology and its Applications*.

- Referee for the *DFG (German Research foundation)*, and the *National Science Center, Poland.*

(Co-)Organization

- **2026:** Working group seminar: Complexity theory, Universidad de Sevilla.
- **Winter term 2021/22 and Summer term 2022:** Learning Seminar: Proof of the topological Poincaré conjecture in dimension 4, following the book *The disk embedding theorem* edited by S. Behren, B. Kalmár, M. Kim, M. Powell, A. Ray, Humboldt University Berlin.
- **19–20 October 2017:** The stabilisation height of fibre surfaces, Lecture series by Filip Misev, University of Cologne.
- **Winter term 2017/18:** Learning Seminar: Computational Topology, following the book *Computational Topology* by H. Edelsbrunner and J. Harer, University of Cologne.
- **17–19 March 2017:** Morse Structures on Open Books, Lecture series by Joan Licata, University of Cologne.
- **Summer term 16:** Learning Seminar: Cobordism theory, following the lecture notes *Notes on Cobordism* by H. Miller, University of Cologne.
- **Winter term 2015/16:** Contact topology graduate student seminar, University of Cologne.
- **Summer term 2015:** Contact topology graduate student seminar, University of Cologne.
- **Summer term 2015:** Fellow Seminar of the Graduiertenkollege 2015, University of Cologne.
- **Winter term 2014/15:** Fellow Seminar of the Graduiertenkollege 2014, University of Cologne.
- **Summer term 2014:** Learning Seminar: Algebraic Geometry, following the book *Algebraic Geometry* by R. Hartshorne, University of Cologne.
- **Winter term 2013/14:** Learning Seminar: Kirby calculus, following part 2 of 4-*Manifolds and Kirby Calculus* by R. Gompf and A. Stipsicz, University of Cologne.
- **Summer term 2013:** Learning Seminar: Characteristic classes II, following the book *Characteristic Classes* by J. Milnor and J. Stasheff, University of Cologne.
- **Winter term 2012/13:** Learning Seminar: Characteristic classes I, following the book *Characteristic Classes* by J. Milnor and J. Stasheff, University of Cologne.

Languages

- German (mother tongue)
- English (fluent)
- Spanish (beginner)

Sevilla, October 24, 2026